Attendance Percentage Calculator

1. Problem Statement

Calculate a student's attendance percentage using the number of classes attended and the total number of classes.

Display whether the student is eligible based on a 75% attendance requirement.

2. Objectives

1. Read student name and attendance details.
2. Calculate attendance percentage.
3. Use an if-else condition to check eligibility.
4. Display a clear attendance report.

3. Assumptions

1. Total classes must be greater than zero.
2. Classes attended cannot be negative or greater than total classes.
3. Minimum required attendance is 75%.

4. Algorithm

1.Start.

2.Read student name, total classes, and classes attended.

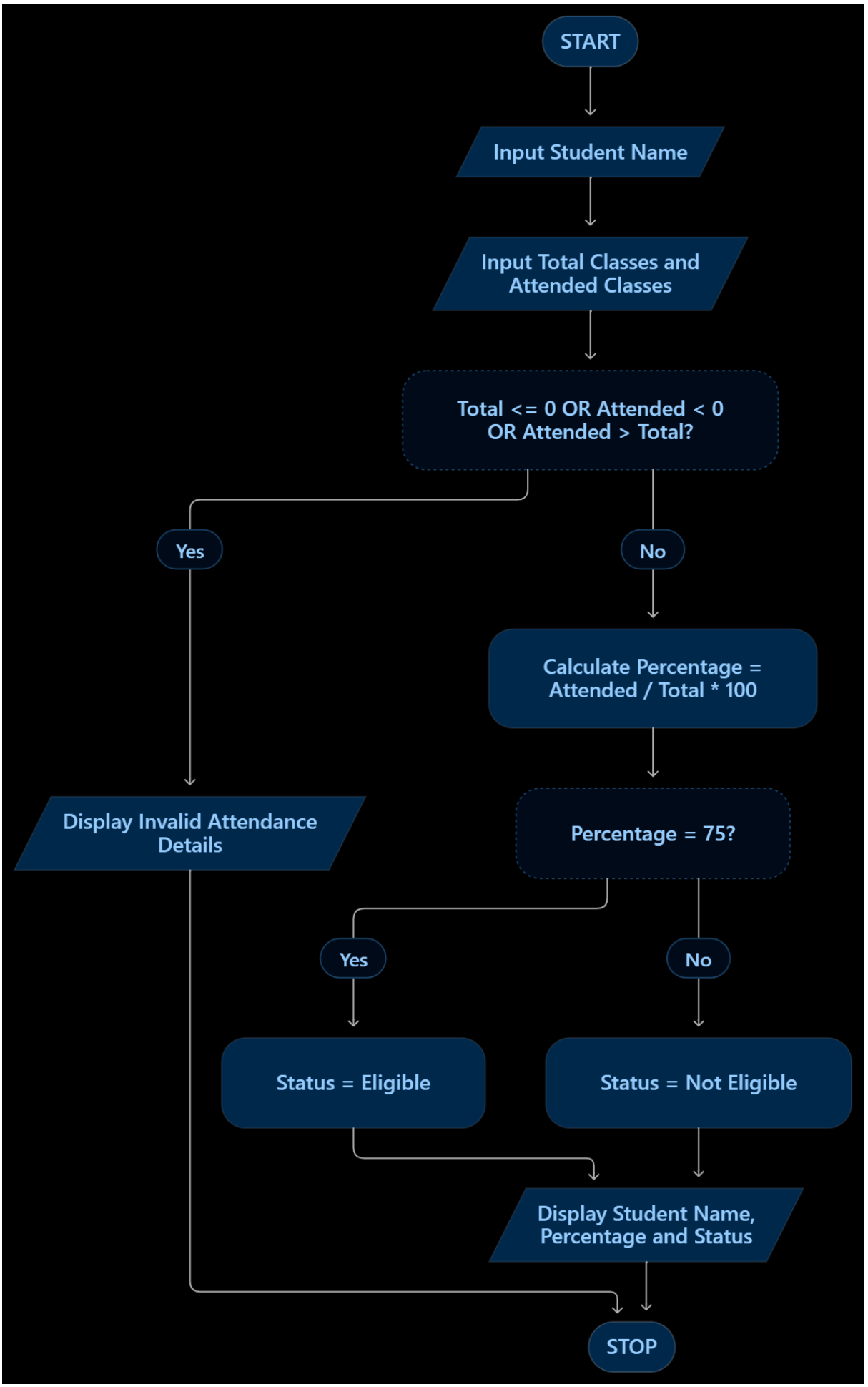
3.Calculate percentage = (attended / total) × 100.

4.If percentage is at least 75, set status to Eligible; otherwise set status to Not Eligible.

5.Display the attendance report.

6.Stop

5. Flowchart


START
Input Student Name
Input Total Classes and Attended Classes
Total <= 0 OR Attended < 0 OR Attended > Total?
Yes
No
Display Invalid Attendance Details
Calculate Percentage = Attended / Total * 100
Percentage = 75?
Yes
No
Status = Eligible
Status = Not Eligible
Display Student Name, Percentage and Status
STOP

6. Pseudocode

```
None
INPUT name, total_classes, attended

percentage = attended / total_classes * 100

IF percentage >= 75 THEN
    status = "Eligible"
ELSE
    status = "Not Eligible"

DISPLAY name, percentage, status
```

8. Test Case and Expected Output

Input:

Input	Value
Name	Ravi
Total classes	100
Attended	82

Calculation:

\[Percentage = \frac{82}{100}\times100 = 82\% \]

Expected Output:

```
None
----- ATTENDANCE REPORT -----
Student: Ravi
Attendance: 82.00%
```

```
Status: Eligible
```

Expected status: Eligible.

9. Conclusion

The program calculates attendance percentage and checks eligibility using a conditional statement.